In solving $\det(A - \lambda I) = 0$, each $\lambda$ satisfying this is an eigenvalue. The number of times that root is repeated in this characteristic equation is called the <u>algebraic multiplicity</u> of that eigenvalue.

The dimension of its associated eigenspace is its <u>geometric multiplicity</u>.

Note that we don't really need $A$ or $\lambda$ to be real to define this, but in most contexts to use these ideas that is needed so we generally assume that.

Very important is that in any such context we have

$$\text{Algebraic multiplicity} \geq \text{Geometric multiplicity} \geq 1$$

<u>Ex:</u> Setting we saw earlier:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{vs.} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Characteristic polynomials: $(1-\lambda)^2$ vs. $(1-\lambda)^2$

So both have $\lambda = 1$ of algebraic multiplicity 2 as only eigenvalue. So then investigate eigenspace for that in each:

$$A - I = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \dashrightarrow \text{eigenspace dimension 2}$$

$$B - I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \dashrightarrow \text{eigenspace dimension 1}$$

So is B diagonalizable? Well it can only be so if it has two linearly independent eigenvectors. But its only only eigenvalue has eigenspace of dimension 1 so can only provide one such eigenvector. Thus B does not have two linearly independent eigenvectors so is not diagonalizable.

In contrast, for $A$ the eigenspace for $\lambda = 1$ has dimension 2, so it will provide two linearly independent vectors that we need so $A$ is diagonalizable (natural as $A$ is already a diagonal matrix!).

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Generalizing this: for $A$ an $n \times n$ real matrix, $A$ is diagonalizable over $\mathbb{R}$ exactly when all of its eigenvalues are real and for each the geometric multiplicity of the eigenvalue matches its algebraic multiplicity.

This is due to creating the needed $n$ linearly independent eigenvectors to diagonalize $A$.

Another way to say this is that real $n \times n$ matrix $A$ is diagonalizable over $\mathbb{R}$ exactly when the sum of the dimensions of its eigenspaces (only real allowed) is $n$.

And this is because the characteristic polynomial is of degree n so the total of the algebraic multiplicities is n.

So if sum of dimensions of eigenspaces is n, geometric multiplicity of each eigenvalue must match its algebraic multiplicity, as needed.

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Several clarifications we can draw from this. First that a real $n \times n$ matrix which has $n$ distinct real eigenvalues is always diagonalizable over $\mathbb{R}$. Why?

Because the algebraic multiplicity of each eigenvalue is 1 so its geometric multiplicity must be 1 as well (can't be 0, or it wouldn't be an eigenvalue). Thus the geometric multiplicity of each eigenvalue does match the algebraic multiplicity, making the matrix diagonalizable.

Another thing is that two similar matrices always have the same eigenvalues. But two matrices with the same eigenvalues are not guaranteed to be similar, the A and B from earlier we showed last time to be of this type.

But if two matrices have same eigenvalues which are distinct real numbers, then they will be similar as they are similar to the same diagonal matrix and thus similar to each other.

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5.3.13

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ 1 & -2 & 2 \end{bmatrix}, \qquad \lambda = 1, 5$$

Again must be repeated eigenvalue here as we discussed last time. So investigate both:

$$A - I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -1 \\ 1 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

via replacing later two rows by themselves minus/plus the first.

Due to two zero rows, this eigenspace has dimension 2. Then since for $\lambda = 5$ the eigenspace has dimension at least 1, the sum of the dimensions must be the needed 3, ensuring A is diagonalizable.

Let us continue to find P and D. From $A - I$, $x_2$, $x_3$ are free and only relevant equation is

$$x_1 + 2x_2 - x_3 = 0$$

As mentioned earlier, when we only have a single linearly independent eigenvector, we likely just brute force find it, but since here there is more than one, probably best to be safer and to do things more formally.

Since $x_2, x_3$ are free so from that solving for $x_1$ we get, using parametric vector form, our general solution is

$$\begin{bmatrix} -2x_2 + x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Thus $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ are two linearly independent

eigenvectors for $\lambda = 1$. Then for other eigenvalue:

$$A - 5I = \begin{bmatrix} -3 & 2 & -1 \\ 1 & -2 & -1 \\ 1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 0 & 8 & 8 \\ 0 & -4 & -4 \\ -1 & -2 & -3 \end{bmatrix}$$

via replacing top two rows with themselves plus multiple of third. Then scaling top two and interchanging first and third gives us

$$\begin{bmatrix} -1 & -2 & -3 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & -2 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

by replacing third row with itself minus second at end. So two pivots, meaning null space dimension 1 which we find via equations

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned}$$

Here we only need one eigenvector so just pick a reasonable nonzero value for one of the variables:

$$x_3 = 1 \text{ ----> } x_2 = -1 \text{ ----> } x_1 = -1$$

So an eigenvector for $\lambda = 5$ is $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$. Thus we have

$$P = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

as our desired diagonalization.

So despite the repeated eigenvalue, this matrix still managed to be diagonalizable, it very much boils down to the specific situation.

# 5.4 Extending Eigenvalues

Say $V$ is a real vector space and $T: V \dashrightarrow V$ is a linear transformation.

We say $\lambda$ is an eigenvalue of $T$ if there exists nonzero $x$ for which

$$T(x) = \lambda x$$

Same as earlier? Well yes same concept of sending vector to multiple of itself. But in broader context since this is general vector space, might not be able to use matrices, e.g. say V is infinite dimensional.

But while can define this in any vector space, in order to try to use earlier ideas we really generally have to assume $V$ is finite dimensional.

When doing $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ we liked to express this as a matrix transformation. How did we find that matrix? Well assuming we were using the standard basis it was given by

$$\begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix}$$

How to find matrix for $T: V \dashrightarrow V$ when $\dim V = n$?

Thing is $V$ may not be $\mathbb{R}^n$ (maybe it's the column space for some matrix perhaps). As such it may not have an inherent "standard basis".

Ok, it's of dimension $n$ so it's isomorphic to $\mathbb{R}^n$ so we could associate it as such, and use standard basis via that isomorphism, but that may not be convenient to set up. As such we usually default to just picking a basis without making that association and using it. Say we use:

$$\mathcal{B} = \{b_1, b_2, \dots, b_n\}$$

Then the matrix representation for $T$ in this basis, called the $\mathcal{B}$-matrix, is given by

$$\begin{bmatrix} T(b_1) & T(b_2) & \dots & T(b_n) \end{bmatrix} = [T]_{\mathcal{B}}$$

Same style as in $\mathbb{R}^n$, just used standard basis there. Why does this work? That is, why does $[x]_{\mathcal{B}}$ map to $[T(x)]_{\mathcal{B}}$? Say

$$x = r_1 b_1 + r_2 b_2 + \dots + r_n b_n$$

That is,

$$[x]_{\mathcal{B}} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}_{\mathcal{B}}$$

Then

$$T([x]_{\mathcal{B}}) = \begin{bmatrix} T(b_1) & T(b_2) & \dots & T(b_n) \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}$$

while

$$[T(x)]_{\mathcal{B}} = \left[ T(r_1 b_1 + r_2 b_2 + \dots + r_n b_n) \right]_{\mathcal{B}}$$

$$= \left[ r_1 T(b_1) + r_2 T(b_2) + \dots + r_n T(b_n) \right]_{\mathcal{B}}$$

via linear

$$= \begin{bmatrix} T(b_1) & T(b_2) & \dots & T(b_n) \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}$$

Note that in this context not big deal whether you write $\mathcal{B}$ or not since that is only basis we are considering, not like in $\mathbb{R}^n$ where we might be worried if we meant standard basis or some special one.

---

Interesting way we use this in a more familiar context:

Say $A$ is an $n \times n$ real matrix and consider
$T: \mathbb{R}^n \dashrightarrow \mathbb{R}^n$ defined by $T(x) = Ax$

We dealt with this type of transformation quite often earlier in class. What if we wanted eigenvectors of this though? Find eigenvalues for $A$? But what if $A$ is a very annoying matrix? Might not want to do that.

Well $A$ is the matrix which represents this transformation in the standard basis. Although this seems unnatural at first since standard basis is usually easier, what if for some basis $\mathcal{B}$ the $\mathcal{B}$-matrix for this transformation was a much easier matrix to work with?

Perhaps a diagonal matrix since that's simplest type.

We talked in Chapter 4 about switching from standard basis to arbitrary basis or vice versa. The motivation to do the latter was natural since standard basis is easier usually. But why do the former? This could be a reason!

So say we tried to find a basis in which the matrix for T, the $\mathcal{B}$-matrix , is a diagonal matrix. Well assuming $A$ is diagonalizable, i.e. $A = PDP^{-1}$ , take $\mathcal{B}$ to be the columns of $P$.

This $\mathcal{B}$ is a basis since $P$ is invertible. What is the $\mathcal{B}$-matrix of $T$? As follows:

$$\left[ T(b_1) \; T(b_2) \ldots T(b_n) \right]_{\mathcal{B}}$$

$$= \left[ Ab_1 \; Ab_2 \; \ldots \; Ab_n \right]_{\mathcal{B}}$$

Note that $P$ converts $\mathcal{B}$-coordinates to standard basis so $P^{-1}$ converts standard basis to $\mathcal{B}$, i.e.

$$\left[ Ab_1 \right]_{\mathcal{B}} = P^{-1} Ab_1$$

So using this idea, the prior expression becomes

$$= \left[ P^{-1}Ab_1 \; P^{-1}Ab_2 \; \ldots \; P^{-1}Ab_n \right]$$

$$= \quad P^{-1} A \begin{bmatrix} b_1 & b_2 & \cdots & b_n \end{bmatrix}$$

$$= \quad P^{-1} AP$$

$$= \quad D$$

So the $\mathcal{B}$-matrix is a diagonal matrix.

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5.4.7

$$A = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Find $\mathcal{B}$-matrix for transformation associated with $A$

$$Ab_1 = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 1b_1 + 0b_2$$

$$Ab_2 = \begin{bmatrix} 3 & 4 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 11 \\ 3 \end{bmatrix} = 5b_1 + 1b_2$$

$$\mathcal{B}\text{-matrix}: \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

But why is this not a diagonal matrix? Wasn't that the intent?

Well picking a specific basis, we might be able to make it such (using a basis made from eigenvectors of $A$), but that doesn't mean it will be diagonal in every basis!

The fact that the basis chosen was seemingly random, no association with $A$, no reason it would become diagonal.

In next example we will try to pick our basis accordingly to make the new matrix diagonal.

5.4.9

$$A = \begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}$$

Here we want basis which makes the transformation associated with A into diagonal matrix.

So we need to diagonalize A:

$$\det(A - \lambda I) = -\lambda(4 - \lambda) + 3 = (\lambda - 1)(\lambda - 3)$$

Thus $\lambda = 1, 3$ are eigenvalues.

$$A - I = \begin{bmatrix} -1 & 1 \\ -3 & 3 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}$$

So $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ eigenvector here. For other

$$A - 3I = \begin{bmatrix} -3 & 1 \\ -3 & 1 \end{bmatrix} \sim \begin{bmatrix} -3 & 1 \\ 0 & 0 \end{bmatrix}$$

Meaning $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ is an eigenvector for this.

So overall:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

is a basis which will make this transformation be represented by a diagonal matrix.

# 5.5 Complex Eigenvalues

For $A$ an $n \times n$ real matrix what if $\det(A - \lambda I)$ has some nonreal roots, i.e. nonreal eigenvalues

Very different setting! We only talked about real eigenvalues, the main issue we faced was if we had repeated eigenvalues.

First, real matrix with any nonreal eigenvalues is never diagonalizable over $\mathbb{R}$ is immediate issue.

Why? Well since $A$ is real, any nonreal eigenvalue can only create nonreal eigenvectors so they cannot contribute to the $n$ eigenvectors needed to make $A$ diagonalizable over $\mathbb{R}$.

And since the real eigenvalues do not have enough algebraic multiplicity to create the full $n$, $A$ is not diagonalizable over $\mathbb{R}$.

So how to break such a matrix down, say we want $A^k$ like we did earlier. Here we can't diagonalize over $\mathbb{R}$.

Could be quite hard, even for 3 x 3, since we will have both real and nonreal eigenvalues. So we mostly just focus on 2 x 2 to keep things within reason.

<u>Ex:</u>

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Characteristic Polynomial: $\lambda^2 + 1$

Nonreal eigenvalues here, can't diagonalize over $\mathbb{R}$

What type of transformation is this (think back to 1.9, a section we largely glossed over, but this is one reason they spent a lot of effort on understanding transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$, is useful here)?

It's a rotation about origin. Logically for nontrivial angles this won't have eigenvectors since no nonzero vector will map to a multiple of itself. Hence logical there would be no real eigenvalues/eigenvectors.

We use rotations as base to study real 2 x 2 matrices with nonreal eigenvalues. Not every matrix of that type is a rotation, but perhaps it's similar to such a rotation? Could use that if so, not so different than associating matrices with distinct real eigenvalues with diagonal matrices.

Staple thing is matrix of type $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$

for real $a, b$, $b$ nonzero, has eigenvalues $a \pm bi$

This can be written as

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

for $a = r\cos\phi$ , $b = r\sin\phi$

ideas of polar coordinates. And the first matrix is scaling, second is rotation which both iterate quite reasonably, hence the motivation to associate with this.

This can be tougher since you have to work with $i$ in the calculations.

But in some ways can be easier since for 2 x 2 with nonreal eigenvalues, the two eigenvalues are complex conjugates, e.g. $a \pm bi$ like before. And since they are closely related, maybe their eigenvectors are too.

Perhaps we don't have to find each eigenvector from scratch as we do with distinct real eigenvalues?

One of several nuances to investigate in this setting for sure.